PSET 4: Linear algebra

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Matrix inversion

Invert each of the following matrices by hand (you can use a calculator or computer to check your solution, but be sure to show your work). Hint: you can verify you have the correct inverse by calculating $\mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$. Not all of the matrices may be invertible—if not, show why.[1]

a. $\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

Problem 2: Dummy encoding for categorical variables

Ordinary least squares regression is a common method for obtaining regression parameters relating a set of explanatory variables with a continuous outcome of interest. The vector $\hat{\mathbf{b}}$ that contains the intercept and the regression slope is calculated by the equation:[2]

$$\hat{\mathbf{b}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

If an explanatory variable is nominal (i.e. ordering does not matter) with more than two classes (e.g. $\{\text{White}, \text{Black}, \text{Asian}, \text{Mixed}, \text{Other}\}$), the variable must be modified to include in the regression model. A common technique known as **dummy encoding** converts the column into a series of $n-1$ binary (0/1) columns where each column represents a single class and n is the total number of unique classes in the original column. Explain why this method converts the column into $n-1$ columns, rather than n columns, in terms of linear algebra. **Reminder: X contains both the dummy encoded columns as well as a column of 1s representing the intercept.**

[1] inspired by Simon and Blume 8.19

[2] inspired by Benjamin Soltoff

Problem 3: Solve the system of equations

Solve the following systems of equations for x, y, z, either via matrix inversion or substitution. If a system cannot be solved this way, explain why and describe the full set of solutions.[3]

a. System #1

$$\begin{aligned} x + y + 2z &= 2 \\ 3x - 2y + z &= 1 \\ y - z &= 3 \end{aligned}$$

b. System #2

$$\begin{aligned} y - x &= 1 \\ z - 2y + x &= 1 \\ 3z - 4x + y &= 10 \end{aligned}$$

Problem 4: Multiplying by 0

When it comes to real numbers, we know that if $xy = 0$, then either $x = 0$ or $y = 0$ or both. One might believe that a similar idea applies to matrices, but one would be wrong. Prove that if the matrix product $\mathbf{AB} = \mathbf{0}$ (by which we mean a matrix of appropriate dimensionality made up entirely of zeroes), then it is not necessarily true that either $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$. Hint: in order to prove that something is not always true, simply identify one example where $\mathbf{AB} = \mathbf{0}$, $\mathbf{A}, \mathbf{B} \neq \mathbf{0}$.[4]

Problem 5: Determinants

Find the determinants of the following matrices:

a. $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$

c. $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Problem 6: Cofactor expansion

Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

a. Find all nine minors M_{ij} and use them to build the cofactor matrix $\mathbf{C}$.

b. Use the first row of $\mathbf{C}$ to compute $\det(\mathbf{A})$ by cofactor expansion.

[3] inspired by Gill 4.19

[4] inspired by Grimmer HW5.5

c. Repeat using the second *column* and confirm you get the same answer.

d. Which calculation was less work, and why?

e. Use $\mathbf{C}$ to find $\mathbf{A^{-1}}$

Problem 7: Eigenvalues and eigenvectors

Using the same matrix $\mathbf{A}$ from the previous problem:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

a. Find the trace of $\mathbf{A}$

b. Verify that $\lambda = -1$ is a root. Factor it out and use the quadratic formula on what remains to find the other two eigenvalues.

c. Find an eigenvector for $\lambda = -1$, and verify that $\mathbf{Av} = -\mathbf{v}$.

d. A matrix is **positive definite** when all of its eigenvalues are positive. Is $\mathbf{A}$ positive definite? Why or why not?

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.